

Student Name:

Exploration 1.4 Competitive Advantage to Uniform Colors?

In this exploration, we are going to explore three factors that influence the strength of evidence in a test of significance: (1) the difference between the observed sample statistic and the value of the parameter used in the null hypothesis; (2) the sample size; and (3) one-sided tests versus two-sided tests. To do this we will look at a study conducted by Hill and Barton (*Nature*, 2005) to investigate whether Olympic athletes in certain uniform colors have an advantage over their competitors. They noticed that competitors in the combat sports of boxing, tae kwon do, Greco-Roman wrestling, and freestyle wrestling are randomly assigned red or blue uniforms. For each match in the 2004 Olympics, they recorded the uniform color of the winner.



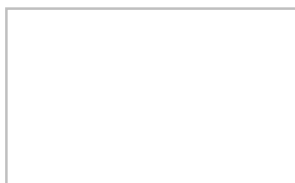
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Think About It

What are the observational units? What is the variable measured? Is the variable categorical or quantitative?

The observational units in this study are the matches, and the variable is whether the match was won by someone wearing red or someone wearing blue—a categorical variable. Let's suppose that going into this study, the researchers wanted to see whether the color red had an advantage over blue. In other words, competitors that wear red uniforms will win a majority of the time.

1. State the null and the alternative hypotheses in words.
2. We will let π represent the probability that a competitor wearing a red uniform wins. Using this, restate the hypotheses using symbols.
3. Researchers Hill and Barton used data collected on the results of 457 matches and found that the competitor wearing red won 248 times, whereas the competitor wearing blue won 209 times. We will carry out a simulation to assess whether or not the observed data provide evidence in support of the research conjecture. This simulation will employ the **3S strategy**: Determine the **statistic**, simulate could-have-been outcomes of the statistic under the null model, and assess the **strength** of evidence against the null model by estimating the p-value or the standardized statistic.
 - a. What is the statistic we will use? Calculate the observed value of the statistic in this study.
 - b. Describe how you could use a coin to develop a null distribution to test our hypothesis.

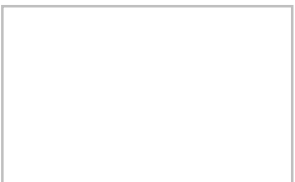


- c. Use the **One Proportion** applet to test our hypothesis. Based on your simulation, find the p-value and write a conclusion. Also write down the mean and standard deviation from your null distribution when the proportion of successes is used for the variable on the horizontal axis. You will need this later.

One-Sided vs. Two-Sided Tests One factor that influences strength of evidence is whether we conduct a two-sided or a one-sided test. Up until now we have only done one-sided tests. In a one-sided test the alternative hypothesis is either $>$ or $<$. In a two-sided test, the alternative hypothesis is $\neq$. So, why would you do a two-sided test and what are the implications?

Suppose the researchers did not necessarily think that red would win more often, but they also didn't necessarily think that blue would win more often. They were just interested in whether one color would win more often than the other. A two-sided alternative hypothesis (red wins at a rate other than 50% of the time, or at a rate not equal to 50%) allows the researchers to be less sure of the anticipated value of the parameter than a one-sided test.

4. If we let π equal the probability that a competitor wearing a red uniform wins, state the hypotheses for this study in symbols using a two-sided alternative.



5. Return to the **One Proportion** applet to approximate the p-value for our original overall proportion of red winning 248 times out of 457 matches, but now select the "Two-sided" check box to find the "two-sided p-value."

- a. Describe how the portion of the null distribution that is shaded red is different than our first test done in #3c.

- b. Describe how the p-value is different from the p-value that was obtained in our original test done in #3c.

- c. To find the two-sided p-value, the applet is looking to see how often 0.543 or larger occurs and how often the comparable value on the other side of the null distribution (or smaller) occurs. To find this value, first compute how far 0.543 is from the center of the null distribution and then go that same distance to the left (less than) the center of the null distribution. What is the comparable value?

- d. Complete the following sentence: The two-sided p-value of _____ is the probability of obtaining _____ or larger plus the probability of obtaining _____ or smaller if the _____ is true.
- e. You should have seen that when the alternative hypothesis is *two-sided*, the p-value is computed by looking at how extreme the observed data is in *both* tails on the null distribution. This makes the p-value about twice as large. Because of this, explain how switching from a one-sided to a two-sided test influences the strength of evidence against the null.

Key Idea

Because the p-value for a two-sided test is about twice as large as that for a one-sided test, they provide less evidence against the null hypothesis. However, two-sided tests are used more often in scientific practice.

Difference Between Statistic and Null Hypothesis Parameter Value

6. A second factor that influences the strength of evidence against the null is how far apart the observed sample statistic and the value of the parameter specified under the null hypothesis are. For this study the null value was 0.50 and the observed sample statistic was about 0.543 (or 54.3% of the competitors wearing red won their matches). Suppose a larger proportion of competitors wearing red won their matches. In fact, suppose 57% of the 457 matches were won by a competitor wearing red.
- a. Go back to the **One Proportion** applet and approximate the (one-sided) p-value for this situation where again we are testing to see whether the overall probability of winning is more than 0.50.
- b. Is your p-value larger or smaller than your original one? Explain why this makes sense.
- c. Write a sentence explaining the relationship between the distance between the observed sample statistic and the value of the parameter specified under the null hypothesis to the strength of evidence against the null hypothesis.



Key Idea

The farther away the observed statistic is from the average value of the null distribution, the more evidence there is against the null hypothesis.

Sample Size

7. The third factor we will look at that influences strength of evidence against the null hypothesis is the sample size. As we said earlier, the data for this study came from four combat sports in the 2004 Olympics. One of those sports was boxing. The researchers found that out of the 272 boxing matches, 150 of them were won by competitors wearing red. This proportion of $150/272 \approx 0.551$ is similar to the overall proportion of times the competitor wearing red won. Let's see what the smaller sample size does to the strength of evidence. Use the **One Proportion** applet to test the same hypotheses as we originally did, but with just the boxing matches as our sample.

- a. Compare the null distribution you generate in this case to that generated in #3. In particular, how do the center (mean) and variability (standard deviation) compare?
- b. What is your new p-value? Is it larger or smaller than your original p-value from #3c? Explain why this makes sense.
- c. Write a sentence explaining the relationship between sample size and the strength of evidence against the null hypothesis.

Key Idea

As the sample size increases (and the value of the observed sample statistic stays the same) the strength of evidence against the null hypothesis increases.